

# St. Vincent Pallotti College of Engineering and Technology, Nagpur

Department of Computer Science and Engineering (Data Science)

ACTIVITY REPORT 2025-2026

|                                                 |                          |
|-------------------------------------------------|--------------------------|
| <b>Title of Project:</b> ATM Transaction System |                          |
| <b>Student Name 1:</b> Saylee Raut              | <b>Roll No:</b> 23011057 |
| <b>Student Name 2:</b> Sharyu Charde            | <b>Roll No:</b> 23011046 |
| <b>Sem:</b> VI                                  |                          |
| <b>Data of Submission:</b> 24-04-26             |                          |

## Objective

The main objective of the **ATM Transaction System** is to develop a user-friendly and secure application that simulates the basic functionalities of an Automated Teller Machine (ATM). The system allows users to perform essential banking operations such as authentication, balance inquiry, deposit, withdrawal, and viewing mini statements.

This project aims to:

- Demonstrate **Object-Oriented Programming (OOP)** concepts in Java
- Provide a **graphical user interface (GUI)** for better user interaction
- Ensure **data validation and security (PIN-based login)**
- Simulate real-world ATM operations in a simplified manner

## Methodology

The project is developed using a **modular and object-oriented approach**. The system is divided into logical components such as user authentication, transaction processing, and GUI handling.

## Steps followed:

1. **Requirement Analysis**  
Identified key ATM functionalities like login, deposit, withdrawal, etc.
2. **System Design**  
Designed classes such as Account and GUI components using Swing.
3. **Implementation**
  - Used Java Swing for GUI
  - Used ArrayList for storing account data
  - Implemented logic for transactions

#### 4. **Testing**

Tested all modules:

- Valid login
- Invalid PIN handling
- Deposit & withdrawal validation

#### 5. **Execution**

Final system runs as a desktop GUI application.

### **Algorithm**

#### **Login Algorithm**

1. Start
  2. Input account number and PIN
  3. Compare with stored data
  4. If match → Login successful
  5. Else → Show error
- 

#### **Deposit Algorithm**

1. Enter amount
  2. Check if amount  $> 0$
  3. Add amount to balance
  4. Update transaction history
  5. Display success message
- 

#### **Withdrawal Algorithm**

1. Enter amount
  2. Check if amount  $\leq$  balance
  3. If yes → deduct amount
  4. Else → show insufficient balance
  5. Update history
-

## Tools & Technologies Used

| Category             | Tool/Technology    | Description                                   |
|----------------------|--------------------|-----------------------------------------------|
| Programming Language | Java               | Core language used for development            |
| GUI Framework        | Java Swing         | Used to design graphical interface            |
| IDE                  | VS Code / IntelliJ | For coding and execution                      |
| Data Storage         | ArrayList          | Used for storing account and transaction data |
| OOP Concepts         | Classes & Objects  | For structured programming                    |
| Platform             | Windows            | Execution environment                         |

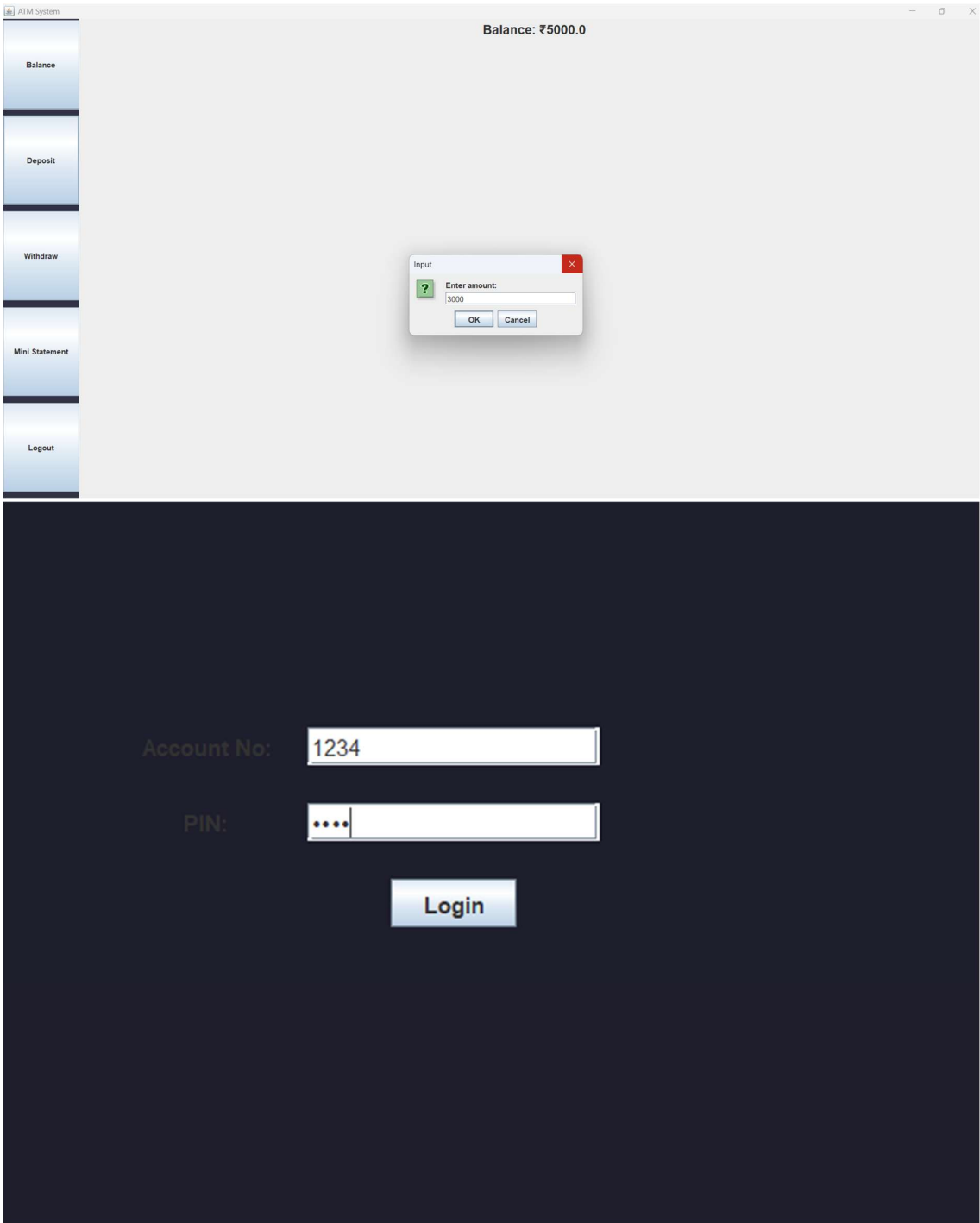
## 5. Main Concept of the Project

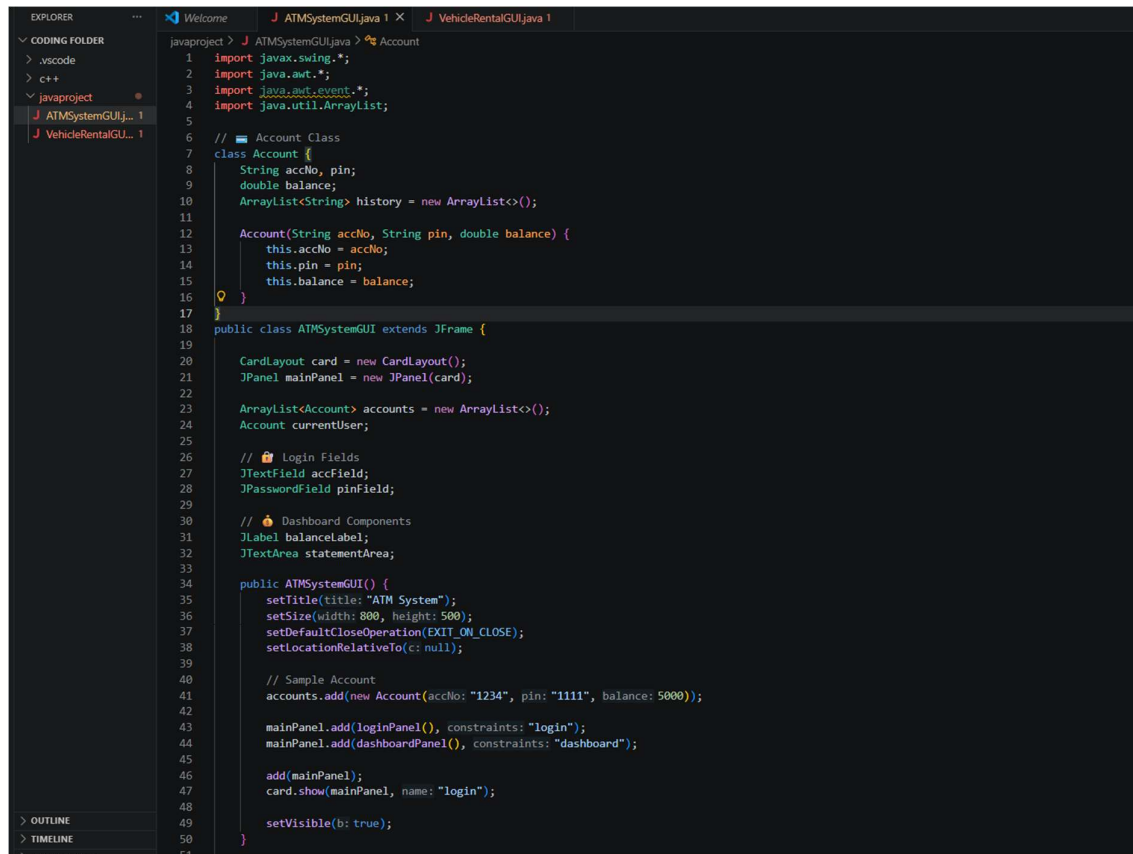
The ATM Transaction System is based on Object-Oriented Programming principles and simulates real-world banking operations.

Core Concepts Used:

- Encapsulation  
Account data (balance, PIN) is stored securely inside class
- Abstraction  
User interacts only with GUI, not internal logic
- Modularity  
Separate methods for deposit, withdraw, login, etc.
- Event-Driven Programming  
GUI buttons trigger actions (like real applications)

Output





```
1 import javax.swing.*;
2 import java.awt.*;
3 import java.awt.event.*;
4 import java.util.ArrayList;
5
6 // Account Class
7 class Account {
8     String accNo, pin;
9     double balance;
10    ArrayList<String> history = new ArrayList<>();
11
12    Account(String accNo, String pin, double balance) {
13        this.accNo = accNo;
14        this.pin = pin;
15        this.balance = balance;
16    }
17 }
18
19 public class ATMSystemGUI extends JFrame {
20
21    CardLayout card = new CardLayout();
22    JPanel mainPanel = new JPanel(card);
23
24    ArrayList<Account> accounts = new ArrayList<>();
25    Account currentUser;
26
27    // Login Fields
28    JTextField accField;
29    JPasswordField pinField;
30
31    // Dashboard Components
32    JLabel balanceLabel;
33    JTextArea statementArea;
34
35    public ATMSystemGUI() {
36        setTitle(title: "ATM System");
37        setSize(width: 800, height: 500);
38        setDefaultCloseOperation(EXIT_ON_CLOSE);
39        setLocationRelativeTo(c: null);
40
41        // Sample Account
42        accounts.add(new Account(accNo: "1234", pin: "1111", balance: 5000));
43
44        mainPanel.add(loginPanel(), constraints: "login");
45        mainPanel.add(dashboardPanel(), constraints: "dashboard");
46
47        add(mainPanel);
48        card.show(mainPanel, name: "login");
49
50        setVisible(b: true);
51    }
52 }
```

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## 7. Result

The ATM Transaction System was successfully designed and implemented using Java Swing. The application provides a user-friendly interface and performs all essential banking operations such as login authentication, deposit, withdrawal, balance inquiry, and mini statement generation.

The system effectively demonstrates the use of OOP concepts, GUI design, and event-driven programming. All functionalities were tested and executed successfully, ensuring correctness and reliability.

This project serves as a strong foundation for developing more advanced banking applications with database integration and enhanced security features.

**Faculty Coordinator:**

**Prof Kalyani Satone**

**Dr Nischal Puri**